Lattice-Offset Control of Mode Localization in a Finite Moiré Nanobeam Cavity

Moohyun Lee
Independent Computational Research Project
moohyun.p.lee@gmail.com

Version 1.0 | August 6, 2026

Abstract

Using a 2D eigensolver and finite-difference time-domain simulations, we demonstrate registry-selectable mode localization in a finite moiré nanobeam cavity. Adjusting the relative offset of the central B row moves the total-electric-field intensity peak along the longitudinal direction of the beam. Across the investigated offset range, the longitudinal peak position changes by $7.94a_A$, whereas the transverse variation is $1.33a_A$. Adjusting the local radius of a transition hole also changes Q and confinement while producing only a small change in resonant frequency. At $R = 52$, the optimized design reaches a two-dimensional $Q \approx 1.03 \times 10^4$. These results indicate that moiré geometry provides an additional degree of freedom for selecting the mode position and confinement during cavity design.

1. Introduction

Moiré structures generate long-period modulation through mismatch between two periodic lattices. This modulation provides an additional degree of freedom for shaping localized optical modes and designing finite cavities [1–4]. In quantum photonics, spatial control of a localized mode may help align the cavity field with a spatially localized emitter. However, in a conventional finite cavity, the mode profile is largely fixed by the initial geometry, and changing its position generally requires structural redesign [5].

This project asks whether the spatial location of a localized mode in a finite moiré nanobeam cavity can be controlled by changing the relative lattice offset. It also examines how the quality factor Q can be tuned using local geometric parameters, particularly the transition-hole radius. The final design is evaluated in terms of normalized frequency, Q, effective mode area, and moiré-cell confinement.

2. System Geometry

2.1 Moiré Nanobeam Geometry

Figure 1 shows the initial design of the moiré cavity. The two A rows contain 13 lattice sites with period a_A, whereas the central B row contains 14 sites with period a_B. The periods satisfy

$$13a_A = 14a_B$$

and therefore $a_B = 13a_A/14$. The relative offset δ of the central B row is varied while the two A rows remain fixed, providing an additional geometric parameter for controlling the localized mode.

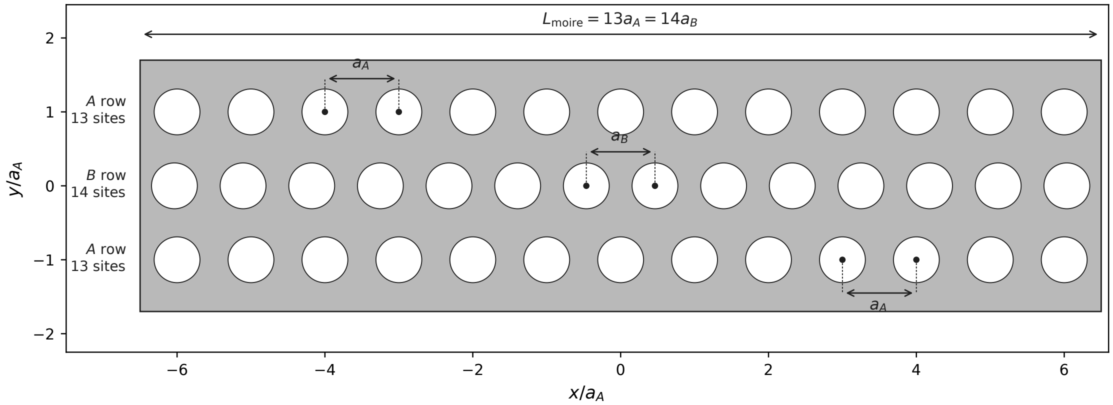

Figure 1: Initial geometry of the commensurate moiré cell. The two A rows contain 13 holes with period a_A, whereas the central B row contains 14 holes with period $a_B = 13a_A/14$.

Table 1: Basic parameters for moiré cavity.

Parameter	Symbol	Value
Relative permittivity of GaAs	ε_r	13
Reference lattice constant	a	$265.85nm$
Lattice-A period	a_A	a
Lattice-B period	a_B	$13a_A/14$
Beam width	w	$3.4a_A$
Relative offset	δ/a_A	-0.3–+0.3
Number of mirror periods	M	15–21
PML thickness	t_{PML}	$1.5a_A$
Spatial resolution	R	32–56

3. Numerical Methods

3.1 MPB Eigenmode Screening

The numerical calculations were performed using the MPB eigensolver and the Meep finite-difference time-domain package [6,7].

The MPB eigensolver was used to identify the photonic band structure and possible bandgap near the target frequency in the ABA moire cell. To investigate the cavity formation mechanism, the representative local configurations, including the AAA, BBB, and ABA cells, were modeled as periodic supercells. Their band structures were compared to determine whether a bandgap or registry-dependent band-edge shift exists near the target frequency.

Based on the simulation results, three candidates, Band 93, 94, and 95, were found within or near the target normalized-frequency range of (0.288100–0.294500), with a target center (0.291613). The Hz field distributions were first used to compare the spatial characteristics and symmetries of the candidate modes.

3.2 Finite-Difference Time-Domain Simulations

The structure was modeled as a two-dimensional, z-invariant system. The spatial resolution R denotes the number of pixels per reference lattice constant a_A. All frequencies are reported in the normalized

Table 2: Additional parameters for moiré cavity.

Parameter	Value
Supercell geometry	$13a_A \times 5a_A$
Commensurate configuration	13–14–13 periods
Boundary conditions	Bloch-periodic
k-points path	$\Gamma \to X$, 42 points
Grid resolution	32
Number of bands	120
Polarization convention	Hz

form $\tilde{f} = fa_A/c = a_A/\lambda$.

An initial broadband FDTD scan showed that the finite-cavity resonance was shifted from the MPB target center of 0.291613. This difference is consistent with the change from a periodic MPB supercell to a finite cavity containing transition and mirror sections. The E_y-polarized source was therefore centered at a normalized frequency of 0.2843 and placed at $(0, 0.3a_A)$ to avoid excitation at a possible symmetry node. The resonances were extracted using Harminv.

3.3 Evaluation Metrics

Quality factor Q

$$Q = -\frac{\omega_r}{2\omega_i} \tag{1}$$

The total electric-field weighting function is defined as

$$u_E(\mathbf{r}) = \varepsilon_r(\mathbf{r}) \sum_\alpha |E_\alpha(\mathbf{r})|^2, \tag{2}$$

where the sum includes all available electric-field components. For the present 2D TE-like simulations, E_x and E_y are used.

The moiré-cell confinement fraction is defined as

$$C_{\mathrm{moire},E} = \frac{\displaystyle\int_{\Omega_{\mathrm{moire}} \cap \Omega_{\mathrm{GaAs}}} u_E(\mathbf{r})\, dA}{\displaystyle\int_{\Omega_{\mathrm{GaAs}}} u_E(\mathbf{r})\, dA}, \tag{3}$$

where the central moiré cell spans $-6.5a_A \le x \le 6.5a_A$, corresponding to one moiré period, $L_{\mathrm{moire}} = 13a_A = 14a_B$.

The normalized two-dimensional effective area is calculated as

$$\frac{A^{(2D)}_{\mathrm{eff},E}}{a_A^2} = \frac{1}{a_A^2} \frac{\displaystyle\int_{\Omega_{\mathrm{GaAs}}} u_E(\mathbf{r})\, dA}{\displaystyle\max_{\mathbf{r}\in\Omega_{\mathrm{GaAs}}} u_E(\mathbf{r})}, \tag{4}$$

where Ω_{GaAs} is identified by $\varepsilon_r(\mathbf{r}) \ge 2.0$.

The frequency is normalized based on the a as

$$\tilde{f} = \frac{fa_A}{c} = \frac{a_A}{\lambda} \tag{5}$$

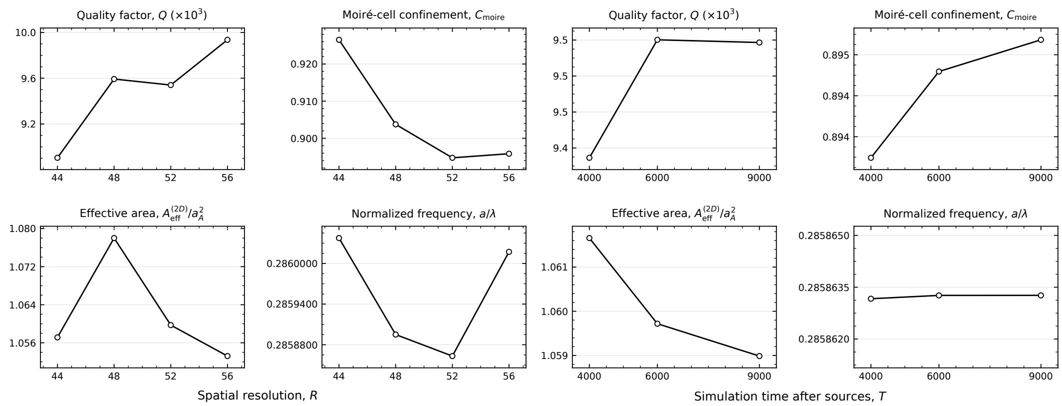

Figure 2: Numerical convergence of Q, normalized frequency, effective mode area, and moiré-cell confinement with respect to spatial resolution R and post-source simulation duration T. The validation was performed at intermediate moiré cavity design right-center hole-radius = 0.308.

4. Results

4.1 Initial Design and Optimization

The initial finite-cavity design without a central-row offset was simulated to identify the target resonance. At $R = 32$, a resonance was found at $f = 0.2846$, with an estimated Q of approximately 1662. To reduce lateral leakage, the BBB mirror sections M were extended, and the hole radius, transition geometry, and central-row offset were optimized. The optimized final design reaches $Q \approx 1.03 \times 10^4$.

4.2 Numerical Convergence

To verify the numerical convergence, the same geometry was simulated at several spatial resolutions and post-source simulation durations. Across $R = 44$–56, the normalized frequency changed by only 0.06%, whereas Q changed by up to 11.6%. The resonant frequency is therefore stable over the investigated range, while Q retains a moderate resolution dependence. The validation was performed at intermediate moiré cavity design right-center hole-radius = 0.308.

Across $T = 4000$–9000, Q changed by approximately 1.03%, while the normalized frequency remained nearly unchanged. These results indicate that the resonance extraction is stable with respect to the post-source simulation duration. A resolution of $R = 52$ was used for the reported final design.

4.3 Offset Dependence

To investigate the effect of the central-row offset, δ/a_A was swept from -0.3 to $+0.3$ in increments of 0.1. The longitudinal peak position x_{peak} moves with the offset, producing a total displacement of $7.94a_A$. In comparison, the transverse peak position varies by only $1.33a_A$. The validation was performed at intermediate moiré cavity design right-center hole-radius = 0.30

The resonant frequency remains within a narrow range throughout the sweep, which is consistent with tracking a closely related resonance. However, Q and C_{moire} decrease sharply at intermediate offsets. The offset therefore provides control over the longitudinal mode position, although this control is accompanied by a trade-off in confinement.

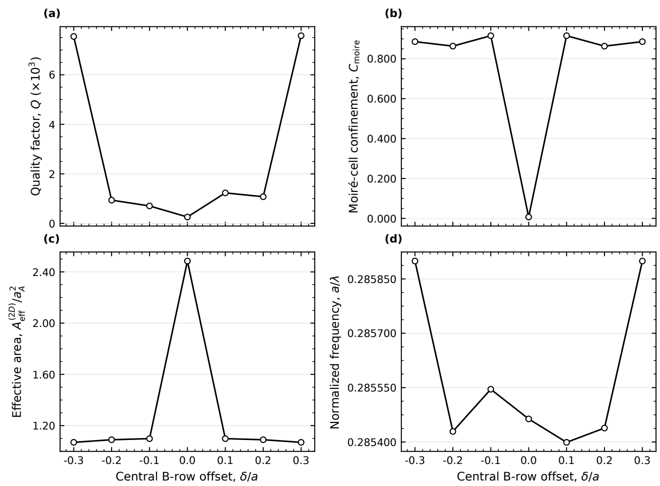

Figure 3: Dependence of (a) Q, (b) C_{moire}, (c) $A_{\mathrm{eff}}^{(2D)}/a_A^2$, and (d) normalized frequency on the central B-row offset δ/a_A. The validation was performed at intermediate moiré cavity design right-center hole-radius = 0.30

Table 3: Parameters of final design.

Parameter	Value
Radius of moiré cell r_A	$0.31a_A$
Radius of moiré cell r_B	$0.31a_A$
δ/a_A	+0.30
Right-center hole-radius in transition column	$0.2985a_A$
Transition radius	$0.339a_A$
Mirror-hole scale	$0.322a_A$
Number of extended mirrors (M)	19
Resolution (R)	52
Run time (T)	6000
Normalized frequency ($\tilde{f}$)	0.285852
Q	1.03×10^4
A_{eff}/a_A^2	1.058
C_moire	0.897

4.4 Transition-Hole Radius Dependence

The transition-hole radius was varied at a fixed offset of $\delta/a = 0.3$ to investigate its effect on cavity leakage. Over the explored range, Q and C_{moire} decrease as the transition-hole radius increases, whereas the normalized frequency changes only slightly. These results indicate that the transition-hole radius primarily controls the confinement and leakage of the localized mode rather than its resonance frequency.

4.5 Final Design

The final design produces a localized field distribution near the selected longitudinal position. The normalized resonant frequency is 0.285852, with $Q = 1.03 \times 10^4$, $A_{\mathrm{eff}}^{2D}/a_A^2 = 1.058$, and $C_{\mathrm{moire}} = 0.897$.

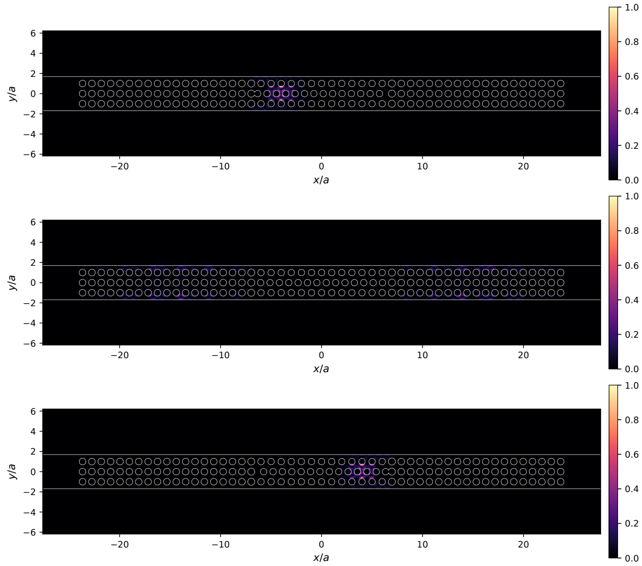

Figure 4: top: $\delta = -0.3$, middle: $\delta = 0$, bottom: $\delta = +0.3$; normalized $\epsilon|E|^2$ profile depending on Offset δ of the center row in moiré cell, by sweeping the values.

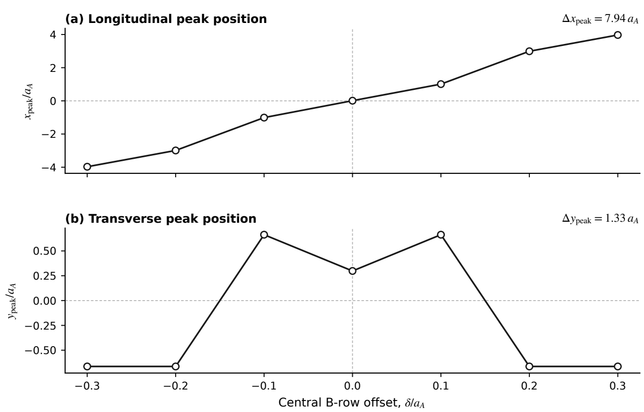

Figure 5: Central B-row offset δ predominantly moves the cavity-mode peak along the longitudinal x direction, with an approximately symmetric displacement about $\delta = 0$. The y_{peak} varied by only $1.33a_A$, compared with a longitudinal shift of $7.94a_A$.

5. Discussion

5.1 Physical Interpretation

The central B-row offset δ is the primary parameter controlling the mode position in this design. Changing the offset modifies the local registry of the moiré cell and selects between approximately mirror-related localized states. Strong confinement is obtained near $|\delta| = 0.3a_A$, whereas intermediate offsets produce substantially lower Q and C_{moire}. The mode is therefore not translated without loss; instead, the offset produces a trade-off between position selection and confinement.

Changing the transition-hole radius modifies the interface between the moiré cell and the surrounding mirror. The resulting variation in Q and C_{moire}, together with the small frequency shift, is consistent with modified interface leakage or mode matching. A directional flux analysis would be required to verify the leakage mechanism directly.

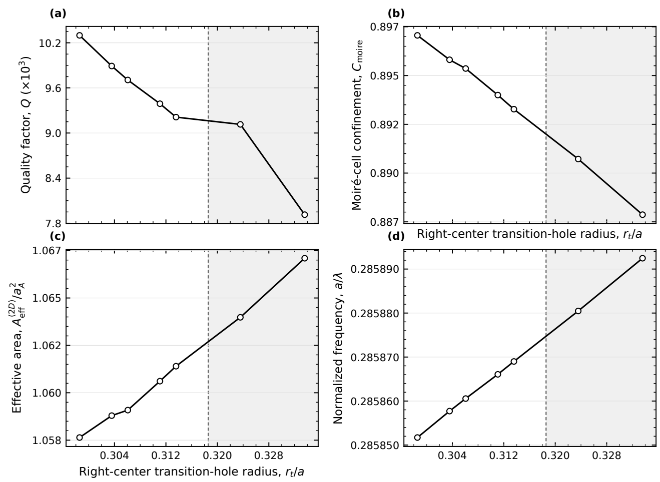

Figure 6: Dependence of (a) Q, (b) C_{moire}, (c) A_{eff}/a_A^2, and (d) normalized frequency on the right-center transition-hole radius r_t/a_A.

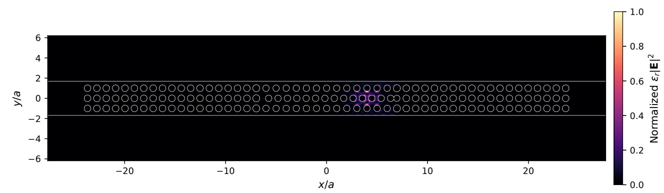

Figure 7: Final design of normalized $\epsilon|E|^2$ profile.

5.2 Claim Boundary

The simulations demonstrate offset-dependent mode localization and geometry-dependent Q tuning within a two-dimensional, z-invariant model. They do not yet establish post-fabrication tunability, fabrication robustness, or enhanced emitter coupling. In addition, the identification of a single mode branch throughout the offset sweep is based on spectral proximity and field-pattern continuity. Quantitative mode-overlap analysis would be required for stricter mode tracking.

6. Limitations and Future Work

Both the MPB eigenmode calculations and the Meep FDTD simulations used two-dimensional, z-invariant models. Consequently, vertical radiation loss, finite slab thickness, substrate effects, and full three-dimensional polarization behavior are not included. The study is also limited by numerical simulations and does not yet account for fabrication disorder or emitter-placement variation. Future work should include full three-dimensional validation, systematic perturbation of the hole positions and radii, and an analysis of emitter-field overlap.

7. Conclusion

Although this study is limited to two-dimensional numerical simulations, it demonstrates that the relative offset of the central B row controls the longitudinal position of a localized mode in a finite moiré nanobeam cavity. Across the investigated offset range, the peak shifts by $7.94a_A$, while its transverse variation remains $1.33a_A$. The transition-hole radius also provides control over Q and confinement with only a small change in resonant frequency, and the optimized design reaches $Q \approx 1.03 \times 10^4$. These results identify the moiré offset as an additional design parameter for controlling mode position and confinement. Full three-dimensional and disorder analyses are required to determine its practical robustness.

8. References

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